AIM :

To write a Python program to solve the 8-Queen problem using the Backtracking technique.

ALGORITHM :

1.Start the program.

2. Create an 8 × 8 chessboard.

3. Place the first queen in the first row.

4. Check whether the position is safe (no queen in the same column or diagonal).

5. If the position is safe, place the queen.

6. Move to the next row and repeat the process.

7. If no safe position is available, backtrack and change the previous queen's position.

8. Repeat until all 8 queens are placed safely.

9. Display the chessboard.

10. Stop the program.

PROGRAM :

```
File  Edit  Format  Run  Options  Window  Help
board = [-1] * 8

def solve(row):
    if row == 8:
        print(board)
        return True

    for col in range(8):
        safe = True

        for i in range(row):
            if board[i] == col or abs(board[i] - col) == abs(i - row):
                safe = False

        if safe:
            board[row] = col
            if solve(row + 1):
                return True

    return False

solve(0)
```

OUTPUT :

```
========== RESTART: C:/Users/nivet/OneDrive/Documents/pgrmm2.py ==========
[0, 4, 7, 5, 2, 6, 1, 3]
>>>
```

RESULT :

Thus, the Python program to solve the 8-Queen problem using the Backtracking technique was executed successfully. All 8 queens were placed on the chessboard without attacking each other.